

Testing Deployment and Maintenance

Construction

Key Definitions

- ❑ **Construction** is the development of all parts of the software itself, documentation, and new operating procedures.
- ❑ **Testing** is a form of insurance. It costs more to repair software bugs when people are depending on the programs than in earlier stages before the systems are in use.
- ❑ **Documentation** provides information to make the system easier to use and repair.

Managing Programming

Main Tasks of Managing the Programming Effort

- ☑ Assigning the programmers
- ☑ Coordinating the activities
- ☑ Managing the schedule

The Programmer Paradox

- ❑ After an appropriate number of people are assigned to a programming task, adding more people slows down rather than speeds up completion of the project.
- ❑ When projects are so complex they require a large team, the best strategy is to break the project into a series of smaller parts that can function as independently as possible.

Coordinating Activities

- ☑ Weekly (hopefully brief) meetings
- ☑ Create and follow standards
- ☑ Separate development, testing, and production areas
 - Separate directories on a hard disk
 - Separate servers
 - Different physical locations
- ☑ Use the program log

Managing the Schedule

- ☑ Use initial time estimates as a baseline
- ☑ Revise time estimates as construction proceeds
- ☑ Fight against scope creep
- ☑ Monitor “minor” slippage
- ☑ Create risk assessment and track changing risks
- ☑ Fight the temptation to lower quality to meet unreasonable schedule demands

Avoid Classic Mistakes

1. Research-oriented development

If you use state-of-the art technology, lengthen planned time

2. Using “low-cost” personnel

If using a significant number of entry level personnel, lengthen planned time

3. Lack of code control

Use source code library to keep programmers from changing the same code at the same time

4. Inadequate testing

Always allocate sufficient time for formal testing

Designing Tests

Testing Philosophy

- ☑ It is dangerous to test early modules without an overall testing plan
- ☑ It may be difficult to reproduce sequence of events causing an error
- ☑ Testing must be done systematically and results documented carefully

Test Plan

Test Plan			Page ____ of ____
Program ID: _____		Version number: _____	
Tester: _____		Date designed: _____	Date conducted: _____
Results: ☐ Passed ☐ Open items: _____			
Test ID: _____		Requirement addressed: _____	
Objective: _____			
Test cases			
Interface ID	Data Field	Value Entered	
1. _____	_____	_____	
2. _____	_____	_____	
3. _____	_____	_____	
4. _____	_____	_____	
5. _____	_____	_____	
6. _____	_____	_____	
Script			

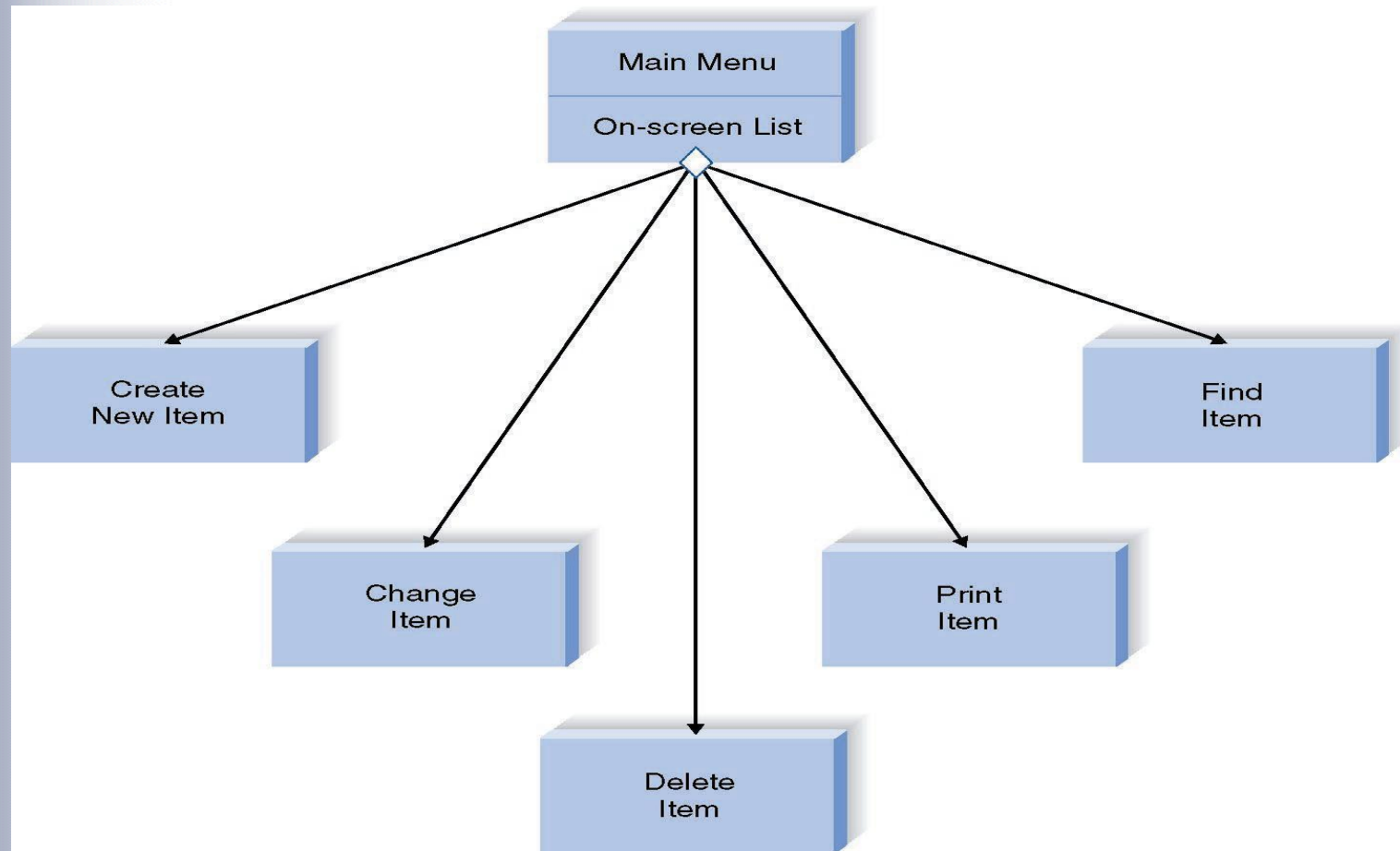
Expected results/notes			

Actual results/notes			

Categories of Testing

- ☑ Stub testing
 - Tests control structures and hardcoded values
 - ☑ Unit testing
 - Tests each module to assure that it performs its function
 - ☑ Integration testing
 - Tests the interaction of modules to assure that they work together
 - ☑ System testing
 - Tests to assure that the software works well as part of the overall system
 - ☑ Acceptance testing
 - Tests to assure that the system serves organizational needs
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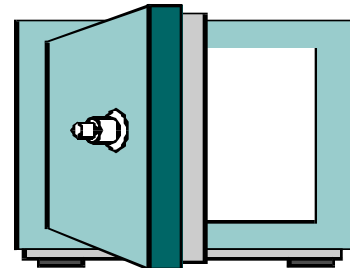
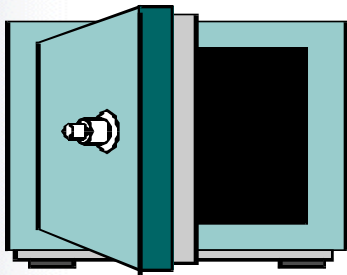
Stub Testing



Unit Testing

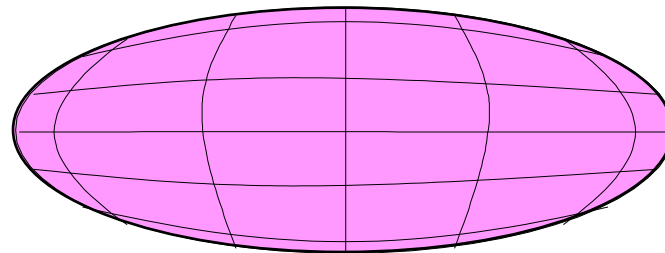
☑ Black Box Testing

- ▣ Focuses on whether the unit meets requirements stated in specification
- ▣ White-Box Testing
- ▣ Looks inside the module to test its major elements



Integration Testing

- ☑ User interface testing
 - Tests each interface function
- ☑ Use-case testing
 - Ensures that each use case works correctly
- ☑ Data flow testing
 - Tests each process in a step-by-step fashion
- ☑ System interface testing
 - Ensures data transfer between systems



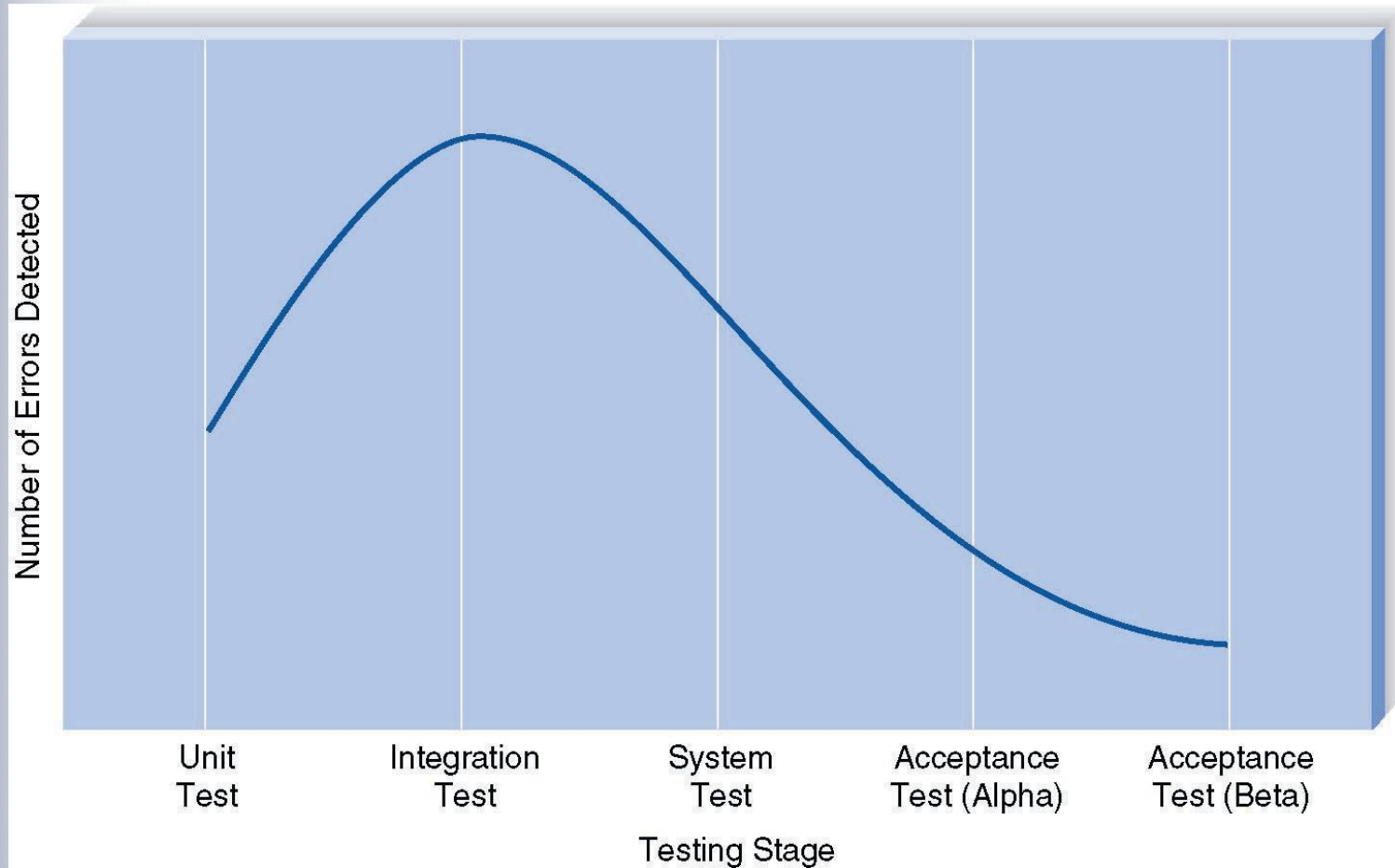
System Testing

- ☑ Requirements/Functional Testing
 - Ensures that integration did not cause new errors
- ☑ Usability Testing
 - Tests how easy and error-free the system is in use
- ☑ Security Testing
 - Assures that security functions are handled properly
- ☑ Performance Testing
 - Assures that the system works under high volumes of activity
- ☑ Documentation Testing
 - Analysts check that documentation and examples work properly

Acceptance Testing

- ☑ Alpha Testing
 - Repeats tests by users to assure they accept the system
- ☑ Beta Testing
 - Actual users monitor for errors or critical improvements

Error Discovery Rates By Test Stage



Your Turn

- ❑ Suppose you were in charge of developing an ATM for a bank
- ❑ What elements would you include in a testing plan before putting the system into operation?

Developing Documentation

Types of Documentation

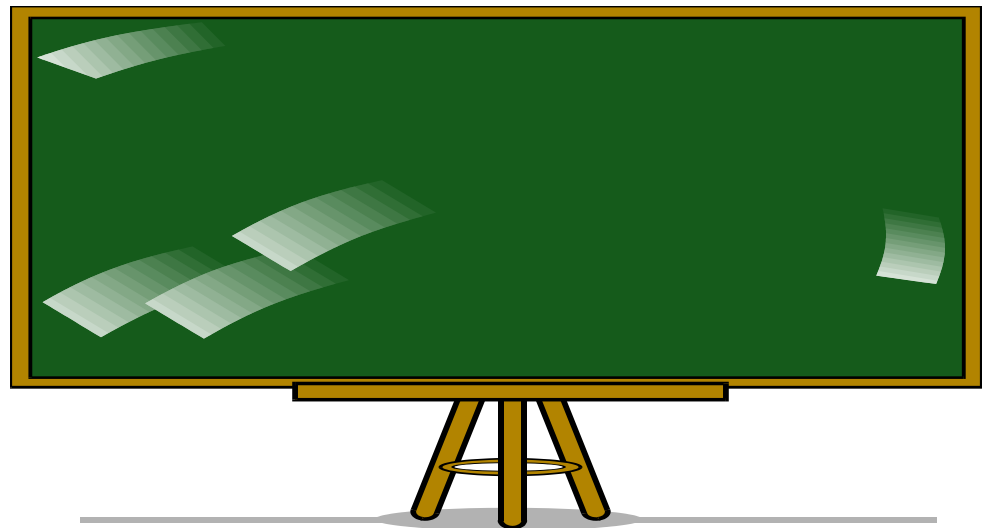
- ☑ System Documentation
 - Intended to help programmers and analysts understand and maintain the system after it is installed
- ☑ User Documentation
 - Intended to help users operate the system

Producing Documentation

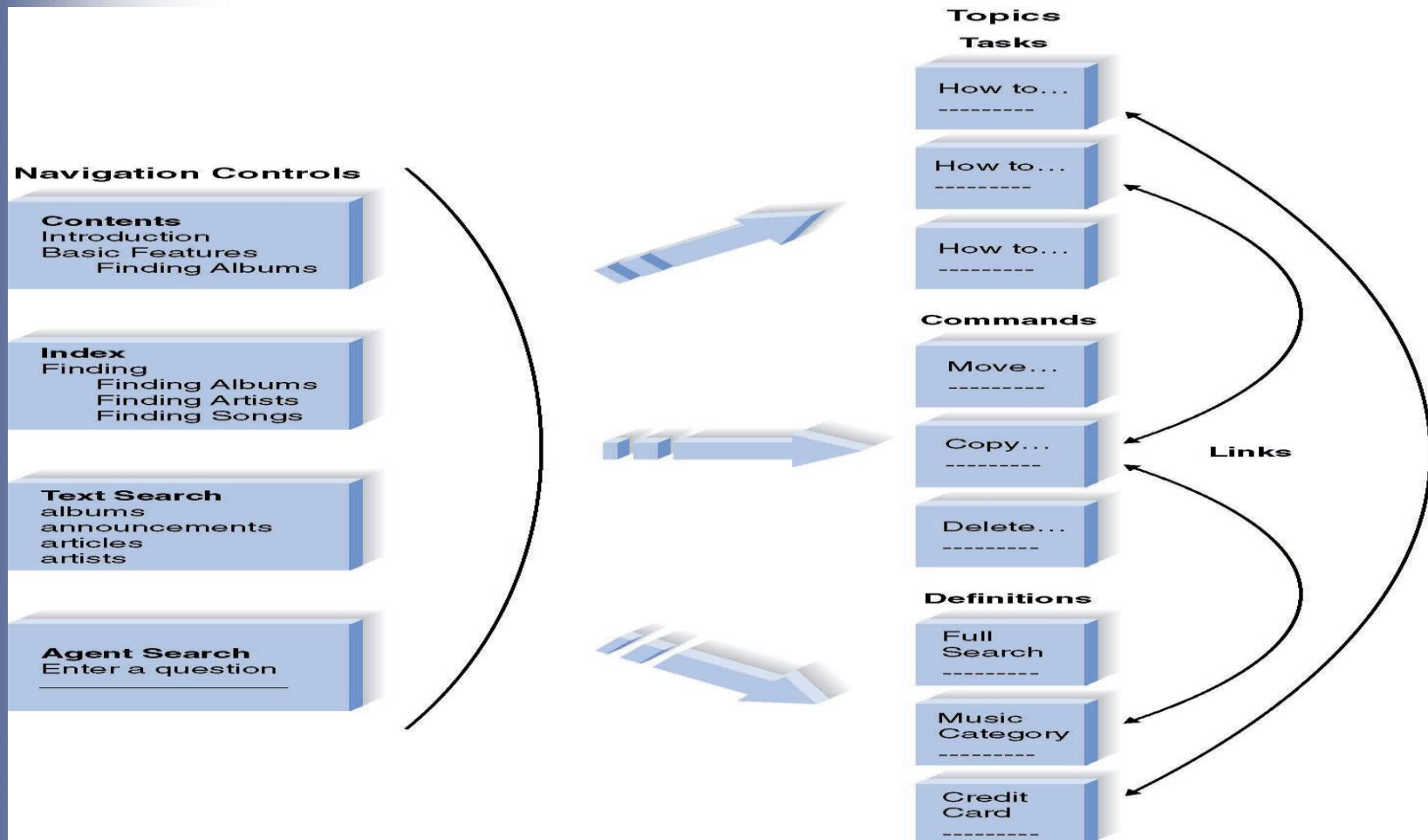
- ☑ High quality documentation takes about 3 hours per page
- ☑ The task should not be left to the end of the project
- ☑ Time required to develop and test user documentation should be built into project plan
- ☑ On-line documentation is growing in importance

Types of User Documentation

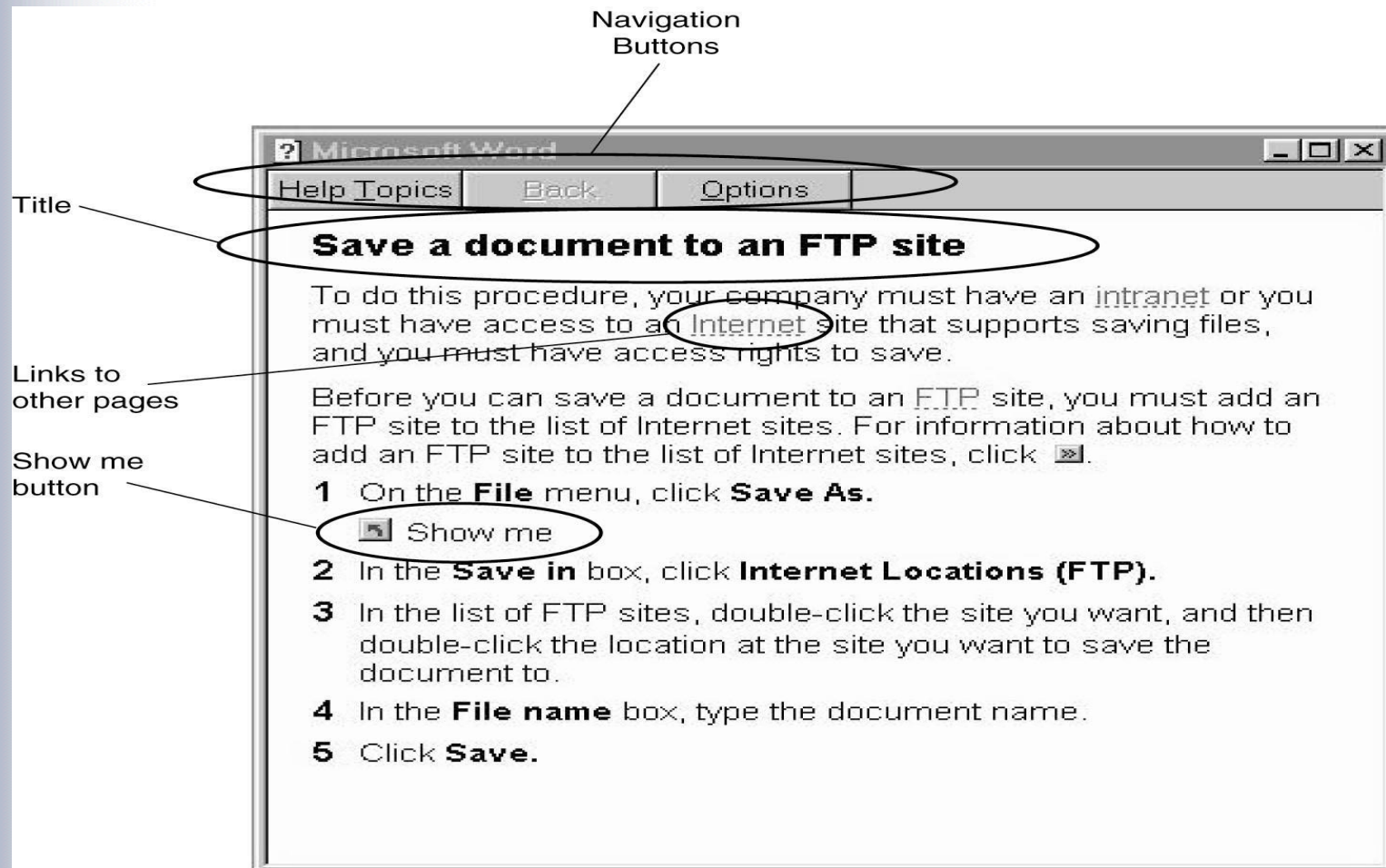
- ☑ Reference documents
- ☑ Procedures manuals
- ☑ Tutorials



Organizing Online Reference Documents



A Help Topic in Microsoft Word



Guidelines for Crafting Documentation Topics

- ☑ Use the active voice
- ☑ Minimize use of “to be” verbs
- ☑ Use consistent terms
- ☑ Use simple language
- ☑ Use friendly language
- ☑ Use parallel grammatical structure
- ☑ Use steps correctly
- ☑ Use short paragraphs

Applying The Concepts At CD Selections

CD Selections

Test Plan

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Program ID: ORD56 Version number: 3

Tester: Smith Date designed: 9/9 Date conducted: 9/9

Results: ☐ Passed ☐ Open items:

Test ID: 12 Requirement addressed: Verify ordering information

Objective:

Ensure that the information entered by the customer on the place-order form is valid

Test cases

Interface ID	Data Field	Value Entered
1. <u>ORD56-3.5</u>	<u>ZIP code/postal code</u>	<u>blank</u>
2. <u>ORD56-3.5</u>	<u>ZIP code/postal code</u>	<u>9021</u>
3. <u>ORD56-3.5</u>	<u>ZIP code/postal code</u>	<u>90210</u>
4. <u>ORD56-3.5</u>	<u>ZIP code/postal code</u>	<u>C1A5B</u>
5. <u>ORD56-3.5</u>	<u>ZIP code/postal code</u>	<u>CAA 2C6</u>
6. <u>ORD56-3.5</u>	<u>ZIP code/postal code</u>	<u>C1A 2C6</u>

Script

Expected results notes

Test 3 and 6 are valid U.S. and Canadian codes that match tested city. All others should be rejected.

Actual results notes

Test 3 and 6 accepted. Tests 1, 2, 4, and 5 were rejected with correct error message.

CD Selections

Help Topic

How to Place an Order

When you are ready to pay for the merchandise you have selected (the items in your shopping cart) you can place your order. There are four steps.

1. Move to the Place order Page

Click on the [Place order](#) button to move to the place order page.

1. Make sure you are ordering what you want

The place order screen displays all the items in your shopping cart. Read through the list to make sure these are what you want because once you submit your credit card information you cannot change the order.

you can delete an item by

Navigation Controls

Table of Contents list:
[How to Place an Order](#)

Index list:
[Credit Card](#)
[Order](#)
[Pay](#)
[Place order](#)

Search find by:
[Credit Card](#)
[Delete Items](#)
[Order](#)
[Pay](#)
[Place order](#)
[Shopping Cart](#)
[Verify Order](#)

Links:
[Shopping Cart](#)

Summary

- ❑ The project manager must assign tasks to programmers, assure coordination of program development and schedule code production and adjust the schedule as development continues.
- ❑ Test planning must be done carefully and a variety of methods all contribute to developing quality software.
- ❑ Documentation should begin well before the completion of coding and testing and address needs of maintenance programmers and system users.

Testing Deployment and Maintenance

Project Initiation

Key Ideas

- ☑ Projects initiation begin when someone sees an opportunity to create business value from using information technology.
- ☑ Feasibility analysis is used to aid in the decision of whether or not to proceed with the IS project.

Key Ideas

- ❑ The **project sponsor** is a key person proposing development or adoption of the new information technology .
- ❑ The **approval committee** reviews proposals from various groups and units in the organization

IDENTIFYING BUSINESS VALUE

Systems Request

- ☑ Key Elements of the Project
 - ▣ Project Name
 - ▣ Project Sponsors
 - ▣ Purpose of the Project
 - ▣ Background of the Project
 - ▣ Scope of the Project
 - ▣ Estimated Project Costs
 - ▣ Expected Date of Completion
 - ▣ Job Assignments
-

FEASIBILITY ANALYSIS

- guides the organization in determining whether to proceed to a project or not.

Identifies the important risk associated with the project that must be addressed if it is approved

Feasibility Analysis

- ☑ Detailing Expected Costs and Benefits
 - ▣ Technical feasibility
 - ▣ Economic feasibility
 - ▣ Organizational feasibility

Technical Feasibility:

Can We Build It?

- ☑ Familiarity with application
 - Knowledge of business domain
 - ☑ Familiarity with technology
 - Extension of existing firm technologies
 - ☑ Project size
 - Number of people, time, and features
-

Economic Feasibility

Should We Build It?

- ☑ Benefits
 - ☑ Development costs
 - ☑ Operational costs
 - ☑ Totals
-

Cash Flow Method for Cost Benefit Analysis

	A	B	C	D	E	F	G
1		Year 1	Year 2	Year 3	Year 4	Year 5	Total
2	Benefits						
3	Tangible	\$\$	\$\$	\$\$	\$\$	\$\$	\$\$\$\$
4	Total Benefits	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$
5	Costs						
6	Development	\$\$	\$\$	\$\$	\$\$	\$\$	\$\$\$\$
7	Operational	\$\$	\$\$	\$\$	\$\$	\$\$	\$\$\$\$
8	Total Costs	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$
9	Benefits - Costs	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$	\$\$\$\$
10							
11							
12							
13							
14							
15							

Return on Investment Calculation

RETURN ON INVESTMENT EQUALS

Total (benefits - costs)

Divided by

Total costs

Net Present Value Calculation

NET PRESENT VALUE EQUALS

Some amount of money

Divided by

$(1 + \text{interest rate})^n$

Where "n" equals the number of periods

Organizational Feasibility

If we build it, will they come?

- ☑ Stakeholder analysis
 - ▣ Is a person, group or organization that can affect a new system
 - ☑ Most Important StakeHolders
 - ▣ Project champion(s)
 - ▣ Organizational management
 - ▣ System users
-

Most Important StakeHolders

- Project champion

Is a high level non-IS executive who is usually but not always the project sponsors who initiated the systems requests

Most Important StakeHolders

- Organizational management
 - Encourage the people in organization to use the system and to accept the many changes that the system will likely create
 - System users
 - Who use the system once it has been installed in the organization
-

Summary

- ☑ **Project initiation** involves creating and assessing goals and expectations for a new system
- ☑ Identifying the **business value** of the new project is a key to success
- ☑ **Feasibility study** is concerned with insuring that technical, economic, and organizational benefits outweigh costs and risks

Testing Deployment and Maintenance

Project Management

Key Definitions

- ☑ **Project management** is the process of planning and controlling the development of a system within a specified timeframe at a minimum cost with the right functionality.
- ☑ A **project manager** has the primary responsibility for managing the hundreds of tasks and roles that need to be carefully coordinated.

CREATING THE WORK PLAN

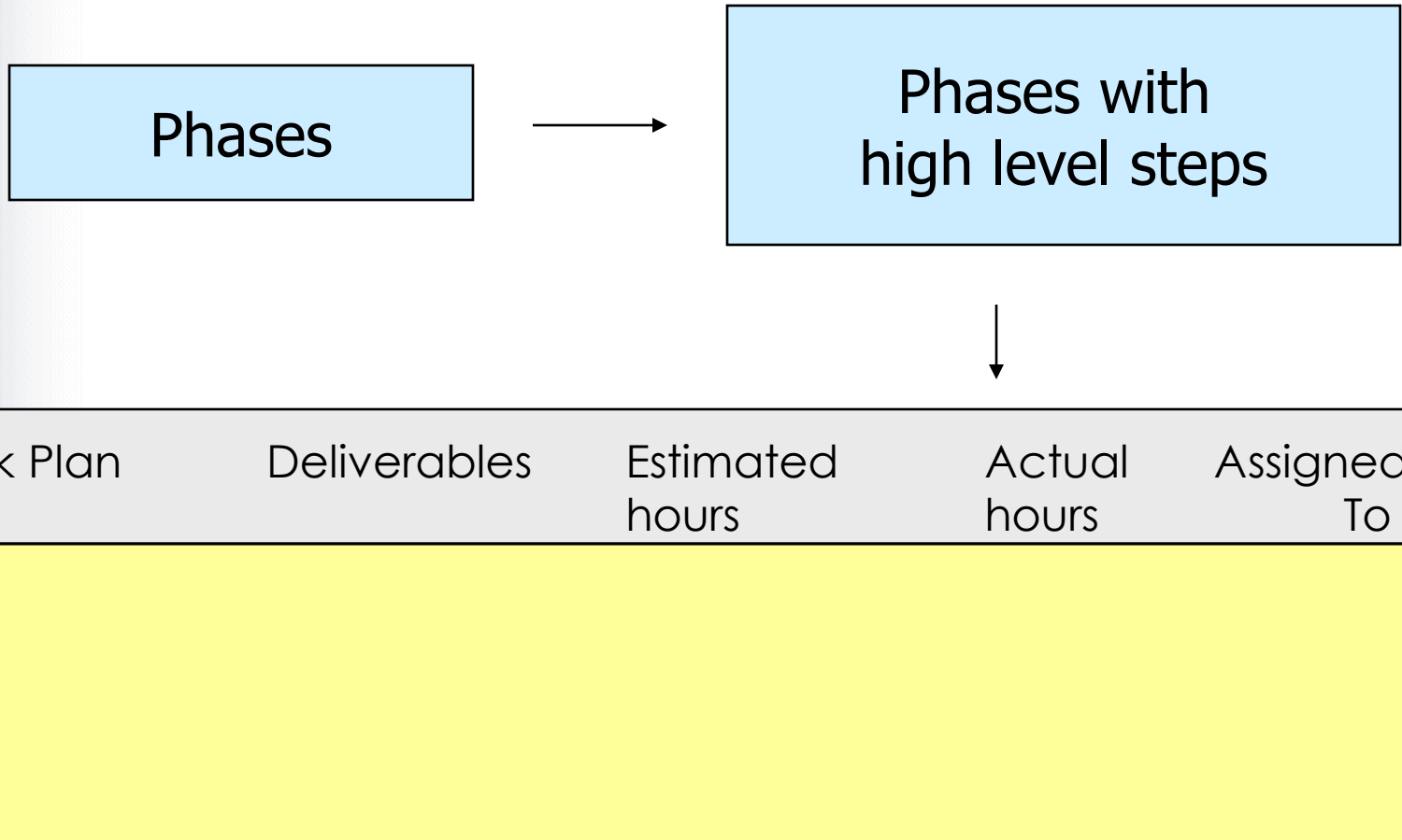
A Workplan Example

Work Plan Information	Example
Name of task Start date Completion date Person assigned Deliverable(s) Completion status Priority Resources needed Estimated time Actual time	Perform economic feasibility Jan 05, 2001 Jan 19, 2001 Mary Smith, sponsor Cost-benefit analysis Open High Spreadsheet 16 hours 14.5 hours

Identifying Tasks

- ☑ Top-down approach
 - Identify highest level tasks
 - Break them into increasingly smaller units
- ☑ Methodology
 - Using standard list of tasks

Top Down Task Identification



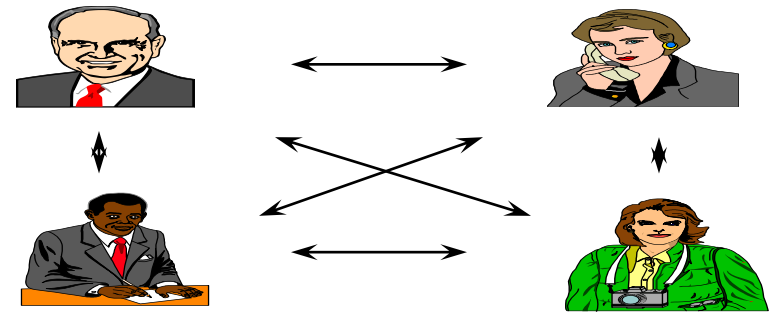
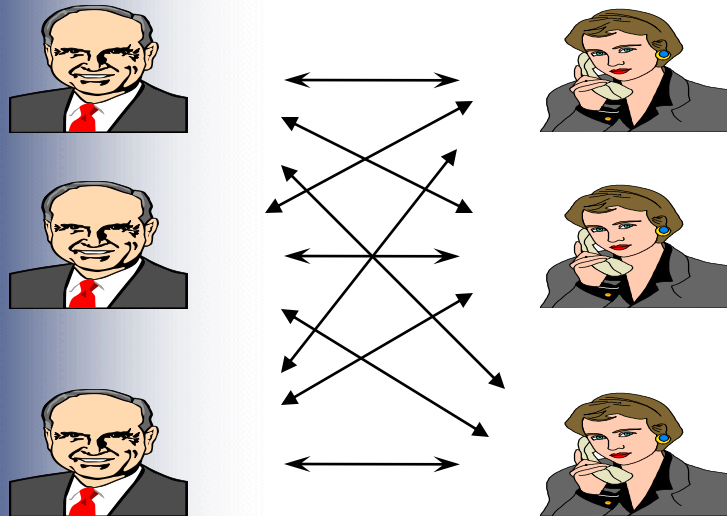
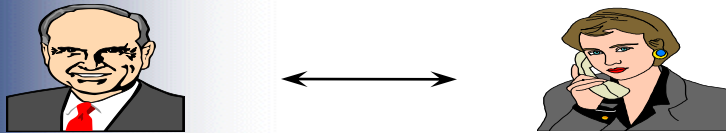
Estimating a Project Based on Industry Information

	Planning	Analysis	Design	Implementation
Industry Standard For Web Applications	15%	20%	35%	30%
Time Required in Person Months	4	5.33	9.33	8

Staffing Attributes

- ☑ Staffing levels will change over a project's lifetime
- ☑ Adding staff may add more overhead than additional labor
- ☑ Using teams of 8-10 reporting in a hierarchical structure can reduce complexity

Increasing Complexity with Larger Teams



Timeboxing

- ☑ Fixed deadline
- ☑ Reduced functionality, if necessary
- ☑ Fewer “finishing touches”

Timeboxing Steps

- ☑ Set delivery date
 - Deadline should not be impossible
 - Should be set by development group
- ☑ Prioritize features by importance
- ☑ Build the system core
- ☑ Postpone unfinished functionality
- ☑ Deliver the system with core functionality
- ☑ Repeat steps 3-5 to add refinements and enhancements

CD Selections

What Should CD Selections' Work Plan Include?

Work Plan	Deliverables	Estimated hours	Actual hours	Assigned to
*				
*				
*				
*				

STAFFING THE PROJECT

Key Definitions

- ☑ The *staffing plan* describes the kinds of people working on the project
- ☑ The *project charter* describes the project's objectives and rules
- ☑ A *functional lead* manages a group of analysts
- ☑ A *technical lead* oversees progress of programmers and technical staff members

Motivation

- ☑ Use monetary rewards cautiously
- ☑ Use intrinsic rewards
 - ▣ Recognition
 - ▣ Achievement
 - ▣ The work itself
 - ▣ Responsibility
 - ▣ Advancement
 - ▣ Chance to learn new skills

Conflict Avoidance Strategies

- ☑ Clearly define roles and project plans
- ☑ Hold individuals accountable
- ☑ Project charter listing norms and groundrules
- ☑ Develop schedule commitments ahead of time
- ☑ Forecast other priorities and their possible impact on the project

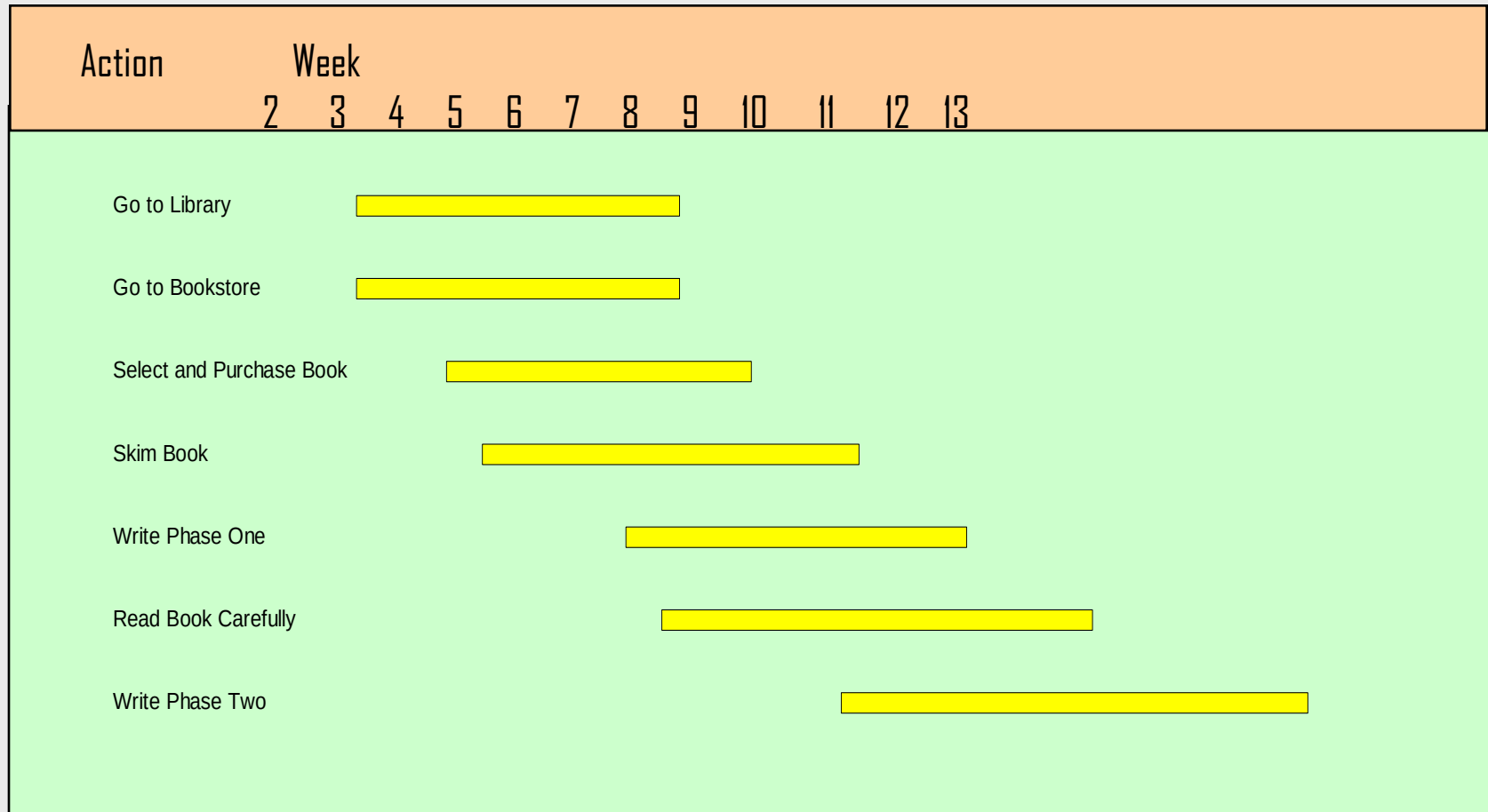
CD Selections

What Should CD Selections' Staffing Plan Include?

Role	Description	Assigned to

CONTROLLING AND DIRECTING THE PROJECT

Tracking Tasks Using Gantt Chart



Summary

- ☑ Project management is critical to successful development of new systems
- ☑ Project management involves planning, controlling and reporting on time, labor, and costs.

Research about the ff:

- ☑ Phased Development Methodology
- ☑ The Prototyping Methodology
- ☑ The Throwaway Prototyping Methodology